Weak collapse of finite arithmetic intertwiners and a typed distributional completion

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Abstract

On $\ell^2(\mathbb{N}_{\geq 1})$ and $L^2(\widehat{\mathbb{Z}}, \mu)$, consider the atomic multiplicative shifts and normalized additive-Haar dilation isometries. We prove that any uniformly bounded sequence intertwining the forward shifts for increasing finite prime sets converges weakly to zero whenever the reciprocal sum of the selected primes diverges. Adjoint intertwining is unnecessary for this conclusion. With adjoints it yields an exact valuation-shell support constraint and a sharp bounded-test coefficient bound. We reconstruct the complementary summable-prime star-intertwiner classification, distinguish weak from strong convergence for finite-prime unitaries, and exhibit a distributional completion with an explicit universal property relative to the locally constant test topology. That completion does not extend the atomic Hilbert norm. All proofs are given; exact finite controls are supplementary. Scholarly priority for the specific arithmetic refinements remains unverified.

1 Scope and calibrated operators

The mathematical statements below have status **PROVED LOCAL, PAPER**; they are not claims of formal verification. This paper reconstructs the atomic/Haar results of the separately inspected repository head `e42096ec0186228dd0f748767599e470f2ca2e1d` (PR178), then strengthens its all-prime star-disjointness to forward disjointness and a finite-to-global limit theorem. Its implementation base is PR179, `81684b4b8956f9f5da56c376ba7e4339277eb665`; PR178 is not silently treated as an ancestor. No analytic source form, zeta zero, or positivity assumption is used.

Write $\mathbb{N}_+ = \mathbb{N}_{\geq 1}$, $H_{\mathrm{d}} = \ell^2(\mathbb{N}_+)$ with basis (e_n), and $H_{\mathrm{c}} = L^2(\widehat{\mathbb{Z}}, \mu)$, where μ is normalized *additive* Haar measure. For $a \in \mathbb{N}_+$ define

$$V_a e_n = e_{an}, \qquad U_a f(x) = \sqrt{a}\, \mathbf{1}_{a\widehat{\mathbb{Z}}}(x) f(x/a).$$

Division is used only on $a\widehat{\mathbb{Z}}$. Multiplication $m_a : x \mapsto ax$ is an injective homeomorphism onto this clopen subgroup, and

$$\mu(aA) = a^{-1}\mu(A) \tag{1}$$

for measurable A. Indeed $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$, the subgroup has index a, and normalized Haar uniqueness gives the restricted scaling law. Consequently both operators are isometries, and direct change of variables gives

$$V_a^* e_n = \begin{cases} e_{n/a} & a \mid n, \\ 0 & a \nmid n, \end{cases} \qquad U_a^* g(y) = a^{-1/2} g(ay), \qquad U_a U_a^* = M_{\mathbf{1}_{a\widehat{\mathbb{Z}}}}. \tag{2}$$

These formulas are well defined on almost-everywhere equivalence classes by (1). Each family is multiplicative; distinct prime isometries doubly commute, as follows from $qx \in p\widehat{\mathbb{Z}} \iff x \in p\widehat{\mathbb{Z}}$ for $p \neq q$.

For a set P of primes write

$$\mathcal{M}_P = \{k \geq 1 : \text{all prime factors of } k \text{ belong to } P\}, \qquad C_P = \{r \geq 1 : p \nmid r \text{ for every } p \in P\}.$$

Every $n \geq 1$ factors uniquely as $n = d_P(n)r$, with $d_P(n) \in \mathcal{M}_P$ and $r \in C_P$. For finite F put

$$E_F = \bigcap_{p \in F} (\widehat{\mathbb{Z}} \setminus p\widehat{\mathbb{Z}}), \quad \alpha_F = \prod_{p \in F} (1 - p^{-1}), \quad d_F(n) = \prod_{p \in F} p^{v_p(n)}, \quad K_{F,n} = d_F(n) E_F.$$

Empty products are one. CRT and (1) give

$$K_{F,n} = \{x : v_p(x) = v_p(n) \ (p \in F)\}, \qquad \mu(K_{F,n}) = \frac{\alpha_F}{d_F(n)}. \tag{3}$$

Here v_p is the valuation of the p-adic coordinate; its possible value $+\infty$ does not satisfy any of the finite equalities in this display.

2 The finite-to-global theorem needs only forward intertwining

Lemma 1 (Finite backward eigenvector calculation)**.** *Let G be a finite prime set, and let $h \in H_{\mathrm{d}}$ satisfy $V_p^* h = \lambda_p h$ for $p \in G$, where $|\lambda_p|^2 = 1/p$. Writing h_n for its coordinates, one has*

$$h_{kr} = \lambda_k h_r \quad (k \in \mathcal{M}_G, \ r \in C_G), \qquad \|h\|^2 = \alpha_G^{-1} \sum_{r \in C_G} |h_r|^2,$$

where $\lambda_k = \prod_{p \in G} \lambda_p^{v_p(k)}$. In particular

$$|h_n| \leq \|h\| \sqrt{\alpha_G / d_G(n)}. \tag{4}$$

Proof. The adjoint equation says $h_{pr} = \lambda_p h_r$ for every r. Iterate it and use the unique factorization $n = kr$. Nonnegative summation and the finite geometric product give

$$\sum_{k \in \mathcal{M}_G} |\lambda_k|^2 = \prod_{p \in G} \sum_{j \geq 0} p^{-j} = \alpha_G^{-1}.$$

This proves the norm identity. The r-summand corresponding to $r = n/d_G(n)$ gives (4). □

Let $\mathcal{D}_m \subset H_{\mathrm{c}}$ be the m-dimensional space of functions depending only on $x \bmod m$.

Theorem 2 (Forward-only cylinder estimate and weak collapse)**.** *Suppose a bounded $T : H_{\mathrm{d}} \to H_{\mathrm{c}}$ satisfies $TV_p = U_p T$ for p in a finite set F. For $g \in \mathcal{D}_m$, put $G = \{p \in F : p \nmid m\}$. Then*

$$|\langle T e_n, g \rangle| \leq \|T\| \sqrt{\frac{m \alpha_G}{d_G(n)}} \, \|g\|_2. \tag{5}$$

If g is a joint eigenvector for the multiplication permutations $g(x) \mapsto g(px)$ on $\mathcal{D}_m$, for $p \in G$, the factor $\sqrt{m}$ can be omitted.

Let $F_1 \subset F_2 \subset \cdots$ be finite sets with union P and $\sum_{p \in P} p^{-1} = \infty$. If

$$\sup_j \|T_j\| < \infty, \qquad T_j V_p = U_p T_j \quad (p \in F_j),$$

then $T_j \longrightarrow 0$ in the weak operator topology. In particular every bounded global forward intertwiner for P is zero.

Proof. For $p \nmid m$, the operator $R_p g(x) = g(px)$ on $\mathcal{D}_m$ permutes the residue classes and hence is unitary for the normalized counting inner product. These operators commute. Finite-dimensional simultaneous diagonalization gives an orthonormal joint eigenbasis $g_1, \dots, g_m$. For each eigenvector g_ℓ, $U_p^* g_\ell = p^{-1/2}\omega_{p,\ell} g_\ell$ with $|\omega_{p,\ell}| = 1$. Taking the adjoint of the *forward* intertwining equation gives

$$V_p^* T^* g_\ell = T^* U_p^* g_\ell = p^{-1/2}\omega_{p,\ell} T^* g_\ell.$$

Apply Lemma 1 to $h = T^* g_\ell$, whose norm is at most $\|T\|$. For $g = \sum_\ell c_\ell g_\ell$, the triangle inequality and $\sum_\ell |c_\ell| \le \sqrt{m}\,\|g\|_2$ give (5). No equation $TV_p^* = U_p^* T$ was assumed.

For a fixed m, removing the finitely many primes dividing m leaves the reciprocal sum divergent. Since $-\log(1 - 1/p) \ge 1/p$, the corresponding products α_{G_j} converge to zero. Thus (5) proves convergence for every fixed e_n and every cylinder test g. The finite linear span of the e_n is dense in H_{d}, and $\bigcup_m \mathcal{D}_m$ is dense in H_{c}: congruence cylinders generate the profinite Borel sigma algebra and their simple functions are dense in L^2. The common norm bound extends the conclusion first to all $g \in H_{\mathrm{c}}$, then to all $f \in H_{\mathrm{d}}$. Explicitly the error from replacing (f, g) by (f_0, g_0) is at most $M\|f - f_0\|\|g\| + M\|f_0\|\|g - g_0\|$, where $M = \sup_j \|T_j\|$. Finally a global intertwiner gives a constant sequence, so it must vanish. □

Remark 3 (What was strengthened). The star-intertwiner obstruction in PR178 remains true, but its adjoint hypothesis is not necessary for atomic-to-Haar bounded disjointness. Adjoints have a different role: they yield the following exact support constraint for arbitrary Haar tests. No forward-only classification in the summable regime is inferred from a star-intertwiner classification.

3 Sharp support under star intertwining

Theorem 4 (Exact shell constraint)**.** *Suppose $T : H_{\mathrm{d}} \to H_{\mathrm{c}}$ is bounded and $TV_p = U_p T$, $TV_p^* = U_p^* T$ for every p in a finite prime set F. For every $n \ge 1$, Te_n vanishes almost everywhere off $K_{F,n}$, and*

$$|\langle Te_n, g\rangle| \le \|T\|\,\|\mathbf{1}_{K_{F,n}} g\|_2 \qquad (g \in L^2(\widehat{\mathbb{Z}}, \mu)), \tag{6}$$

$$|\langle Te_n, g\rangle| \le \|T\|\,\|g\|_\infty \sqrt{\alpha_F / d_F(n)} \qquad (g \in L^\infty(\widehat{\mathbb{Z}}, \mu)). \tag{7}$$

The constant in (7) *is optimal for every F, n. If P is all primes, then for each fixed n and all sufficiently large j, $d_{F_j}(n) = n$, yielding the factor $\sqrt{\alpha_{F_j}/n}$. For a proper prime subset one must retain $d_{F_j}(n)$.*

Proof. Put $d = d_F(n)$ and $r = n/d$. Since $r \in C_F$, $V_p^* e_r = 0$ for all $p \in F$. Therefore $U_p^* T e_r = 0$. By (2), the joint kernel is exactly $L^2(E_F)$, so Te_r is supported on E_F. Forward intertwining gives $Te_n = U_d Te_r$, supported on dE_F. Cauchy–Schwarz, $\|Te_n\| \le \|T\|$, and (3) prove both estimates.

For optimality use the orthogonal shell decompositions established in Proposition 5 (whose proof is independent of this estimate). On the atomic wandering space send e_r to $\alpha_F^{-1/2}\mathbf{1}_{E_F}$ and every other basis vector to zero, and extend by the same selected shifts. This gives a norm-one star intertwiner S satisfying

$$Se_n = \sqrt{d/\alpha_F}\,\mathbf{1}_{dE_F}.$$

The test $g = 1$ attains equality in (7). □

4 The summable complement and the topology boundary

For an arbitrary prime set P define $E_P = \bigcap_{p\in P}(\widehat{\mathbb{Z}} \setminus p\widehat{\mathbb{Z}})$ and $\alpha_P = \mu(E_P) = \prod_{p\in P}(1-1/p)$, by continuity from above.

Proposition 5 (Reconstructed sharp star-intertwiner classification)**.** *The following conditions are equivalent:*

1. $\sum_{p\in P} 1/p < \infty$*;*
2. $\alpha_P > 0$*;*
3. *the restricted atomic and Haar tuples are unitarily equivalent;*
4. *a nonzero bounded star intertwiner $H_{\mathrm{d}} \to H_{\mathrm{c}}$ exists.*

In this regime put $W_{\mathrm{d}} = \ell^2(C_P)$ and $W_{\mathrm{c}} = L^2(E_P, \mu)$. There are orthogonal decompositions

$$H_{\mathrm{d}} = \bigoplus_{k\in\mathcal{M}_P} V_k W_{\mathrm{d}}, \qquad H_{\mathrm{c}} = \bigoplus_{k\in\mathcal{M}_P} U_k W_{\mathrm{c}}. \tag{8}$$

Relative to these identifications, all bounded star intertwiners are $I_{\ell^2(\mathcal{M}_P)} \otimes B$, for arbitrary bounded $B : W_{\mathrm{d}} \to W_{\mathrm{c}}$, with the same norm.

Proof. For $p \geq 2$, $1/p \leq -\log(1-1/p) \leq 2/p$; this proves (1)⇔(2). If the sum diverges, Theorem 2 excludes even a nonzero bounded forward intertwiner.

Assume $\alpha_P > 0$. Unique integer factorization proves the atomic decomposition. The Haar sets kE_P, $k \in \mathcal{M}_P$, are pairwise disjoint, since their selected valuations specify k. They have masses α_P/k. Monotone finite Euler products give

$$\sum_{k\in\mathcal{M}_P} \frac{1}{k} = \lim_{F\uparrow P} \prod_{p\in F} (1-1/p)^{-1} = \alpha_P^{-1}.$$

Thus these sets cover Haar measure one. The isometry U_k maps $L^2(E_P)$ onto $L^2(kE_P)$, proving the Haar decomposition.

The complement of P is infinite, because the reciprocal sum over all primes diverges. In particular C_P is countably infinite. Also E_P leaves the $\mathbb{Z}_q$ coordinate unrestricted for any $q \notin P$; hence its positive finite measure is nonatomic and its L^2 space is infinite-dimensional and separable. Thus the two wandering spaces are unitarily equivalent. Any bounded B extends by $T(V_k w) = U_k B w$ on the orthogonal sums; norm equality and both intertwining equations follow directly, including the lowering at exponent zero. A unitary B yields a unitary T.

Conversely a star intertwiner sends the joint adjoint kernel W_{d} into W_{c}; forward intertwining determines its values on every summand in (8). This proves the classification and the remaining implications. □

Corollary 6 (Finite unitaries collapse only weakly)**.** *For every finite F a unitary star intertwiner exists. If F_j exhausts a prime set with divergent reciprocal sum,* every *choice of such unitaries T_j converges WOT to zero. It cannot have a strongly convergent subsequence, and cannot converge in operator norm.*

Proof. Use Proposition 5 and Theorem 2. A strong subsequential limit would also be the weak limit zero, but $\|T_j f\| = \|f\|$ for every f. This is impossible for nonzero f. Norm convergence implies strong convergence. □

Proposition 7 (All possible weak limits and failures of hypotheses)**.** *For increasing F_j with union P, every WOT cluster point of a uniformly bounded family of finite-F_j star intertwiners is a global P star intertwiner. Conversely every bounded global star intertwiner is the limit of a constant such family. Thus, for a prescribed norm bound M, the possible cluster points across all such families are exactly the global star intertwiners of norm at most M.*

Uniform boundedness cannot be omitted from the weak-collapse theorem. Reciprocal divergence cannot be omitted even for unitary star intertwiners. Adjoints cannot be omitted from the exact shell support conclusion, although they can be omitted from weak collapse.

Proof. Multiplication on either side by a fixed bounded operator is WOT continuous. Every selected prime belongs to all sufficiently late F_j, so the intertwining equations pass to cluster points. Norm-closed balls are WOT closed. The converse is the constant-family construction. Existence of cluster points follows, if needed, from compactness of the bounded operator ball in WOT; it is metrizable here because both Hilbert spaces are separable.

For the first failure, choose a finite-F star intertwiner R_F whose wandering restriction sends e_1 to $\alpha_F^{-1}\mathbf{1}_{E_F}$ and annihilates all other atomic wandering basis vectors. The construction in Proposition 5 gives $\|R_F\| = \alpha_F^{-1/2}$ and $\langle R_F e_1, 1\rangle = 1$. Along a divergent exhaustion this is not weakly zero.

For the second failure take any summable P, including an infinite sparse choice of distinct primes $p_j > 2^j$. A global unitary T exists, so $T_j = T$ does not collapse. The family $T_j = (-1)^j T$ shows that no convergence is automatic even in that regime.

For the third failure take $F = \{2\}$ and a norm-one star intertwiner S with $Se_1 = \sqrt{2}\,\mathbf{1}_{E_F}$. Then $T = U_2 S$ still intertwines the forward shift V_2, but $Te_1 = 2\mathbf{1}_{2E_F}$ is supported off E_F. Furthermore $U_2^* T e_1 = Se_1 \neq 0 = TV_2^* e_1$, explicitly detecting the missing adjoint relation. □

5 Distributional transport: an exact interface category

Let $\mathcal{D} = \bigcup_{m\geq 1} \mathcal{D}_m$ be the complex locally constant functions on $\widehat{\mathbb{Z}}$, with the locally convex inductive limit topology along $\mathcal{D}_{j!}$. Each stage is finite-dimensional. Every algebraic linear functional is continuous, because its restrictions to the stages are continuous and the inductive limit has that universal property. Equip $\mathcal{D}'$ with the weak dual topology $\sigma(\mathcal{D}', \mathcal{D})$ and use the *bilinear* distribution pairing. The operators U_p and U_p^* preserve $\mathcal{D}$ continuously: each sends a finite cylinder stage into a finite cylinder stage. Their transposes are the continuous weak-dual operators

$$\langle \mathsf{U}_p \eta, \phi\rangle = \langle \eta, U_p^* \phi\rangle, \qquad \langle \mathsf{A}_p \eta, \phi\rangle = \langle \eta, U_p \phi\rangle.$$

The symbol A_p is a transpose action, not a Hilbert adjoint on $\mathcal{D}'$. On Dirac distributions they act as

$$\mathsf{U}_p \delta_x = p^{-1/2}\delta_{px}, \qquad \mathsf{A}_p \delta_x = \begin{cases} \sqrt{p}\,\delta_{x/p} & x \in p\widehat{\mathbb{Z}}, \\ 0 & x \notin p\widehat{\mathbb{Z}}. \end{cases} \tag{9}$$

Proposition 8 (Algebraic vacuum classification)**.** *Linear maps $J_\eta : c_{00}(\mathbb{N}_+) \to \mathcal{D}'$ satisfying $J_\eta V_p = \mathsf{U}_p J_\eta$ and $J_\eta V_p^* = \mathsf{A}_p J_\eta$ for every prime are in bijection with distributions η supported on $\widehat{\mathbb{Z}}^\times$, by*

$$J_\eta e_n = n^{-1/2}(m_n)_* \eta, \qquad \eta = J_\eta e_1. \tag{10}$$

Every nonzero such J_η is injective.

Proof. Forward equations force (10). The lowering equations at e_1 give $\mathsf{A}_p \eta = 0$. The image $U_p\mathcal{D}$ consists of all locally constant tests supported in $p\widehat{\mathbb{Z}}$, up to the harmless scalar $\sqrt{p}$. Hence η

annihilates tests supported there. A locally constant test supported off $\widehat{\mathbb{Z}}^\times$ has compact support covered by finitely many such clopen sets; refining their Boolean partition shows that it too is annihilated. This is precisely the asserted support condition.

Conversely, support on the units makes the lowered image in (10) zero when $p \nmid n$ and makes it $J_\eta e_{n/p}$ when $p \mid n$, by (9) or the same test calculation for general distributions. This proves both equations. For distinct n, the compact sets $n\widehat{\mathbb{Z}}^\times$ are disjoint: their valuations at all primes recover n. A finite collection of disjoint compact sets can be separated by clopen neighborhoods. The pushforward under m_n is injective on distributions, since every test on $\widehat{\mathbb{Z}}$ can be transferred to $n\widehat{\mathbb{Z}}$ and extended by zero. Therefore no nontrivial finite linear combination of the nonzero distributions $(m_n)_*\eta$ can vanish. This proves injectivity. □

Fix the additional choice $\eta = \delta_1$, and write $Je_n = n^{-1/2}\delta_n$. Put on $c_{00}(\mathbb{N}_+)$ the initial locally convex topology τ_J generated by

$$a \longmapsto \left| \sum_n a_n n^{-1/2} \phi(n) \right|, \qquad \phi \in \mathcal{D}. \tag{11}$$

This topology is specified by the chosen tests, not by the atomic norm.

Theorem 9 (Typed completion and universal property)**.** *The map J identifies $(c_{00}(\mathbb{N}_+), \tau_J)$ with a dense linear subspace of the complete Hausdorff locally convex space $(\mathcal{D}', \sigma(\mathcal{D}', \mathcal{D}))$. Thus $\mathcal{D}'$ is its completion. For every complete Hausdorff locally convex space Y and every continuous linear $L : (c_{00}, \tau_J) \to Y$, there is a unique continuous linear $\overline{L} : \mathcal{D}' \to Y$ satisfying $\overline{L}J = L$.*

More particularly, suppose Y carries continuous linear operators B_p, C_p and L satisfies $LV_p = B_pL$, $LV_p^* = C_pL$. Then its unique extension satisfies $\overline{L}\mathsf{U}_p = B_p\overline{L}$ and $\overline{L}\mathsf{A}_p = C_p\overline{L}$.

Proof. Injectivity follows from Proposition 8, or separation of any finite set of distinct integers by residue tests. The initial topology makes J a topological embedding. For density fix $\eta \in \mathcal{D}'$ and a finite family of tests. All those tests lie in one $\mathcal{D}_m$. Put $b_r = \langle \eta, \mathbf{1}_{r+m\widehat{\mathbb{Z}}} \rangle$ for $r = 0, \dots, m-1$, and let

$$a = \sum_{n=1}^{m} \sqrt{n}\, b_{n \bmod m} e_n.$$

Then Ja agrees exactly with η on every test in $\mathcal{D}_m$. It therefore meets every weak neighborhood specified by the chosen finite tests, proving density.

A pointwise Cauchy net of linear functionals on $\mathcal{D}$ has a pointwise limit which is a linear functional. Every such functional is continuous for the stated inductive limit topology, so the limit belongs to $\mathcal{D}'$. This proves weak-dual completeness; the topology is Hausdorff because the tests separate functionals. Equivalently $\mathcal{D}'$ is the closed subspace of $\mathbb{C}^{\mathcal{D}}$ cut out by the linearity equations.

A continuous linear map between topological vector spaces is uniformly continuous for their additive uniformities. Given η, take a Cauchy net $Ja_i \to \eta$; completeness of Y supplies the limit of La_i. Uniform continuity makes that limit independent of the approximating net, and defines a continuous linear extension. Density and Hausdorffness give uniqueness. These are exactly the completion extension properties, not a universal property in a larger undeclared category. Finally all source and target actions are continuous, so the equivariance equations on the dense image of J extend to $\mathcal{D}'$. □

Proposition 10 (Why this completion cannot preserve the Hilbert interface)**.** *There is no continuous extension of J from the unweighted Hilbert space H_{d} to weak $\mathcal{D}'$. There is no continuous*

linear map $R : \mathcal{D}' \to H_{\mathrm{d}}$ satisfying $RJe_n = e_n$ for every n. More generally, every continuous linear map from (c_{00}, τ_J) into a normed space factors through one finite cylinder projection $a \mapsto Ja|_{\mathcal{D}_m}$, and therefore has finite rank. Thus Theorem 9 does not supply a universal carrier preserving both all locally constant tests and the atomic Hilbert norm.

Proof. Evaluation at the constant test 1 would make $a \mapsto \sum_n a_n/\sqrt{n}$ a bounded functional on ℓ^2. But for $H_N = \sum_{n=1}^{N} 1/n$ and $a^{(N)} = H_N^{-1/2} \sum_{n=1}^{N} n^{-1/2} e_n$, one has $\|a^{(N)}\| = 1$ and $\langle Ja^{(N)}, 1\rangle = \sqrt{H_N} \to \infty$. For the second claim, $Je_n \to 0$ weakly in $\mathcal{D}'$ since $|\langle Je_n, \phi\rangle| \le \|\phi\|_\infty/\sqrt{n}$ for every locally constant ϕ. A continuous R would force $e_n \to 0$ in Hilbert norm, a contradiction. For the last assertion, continuity at zero bounds the norm of La by a constant times the maximum of finitely many defining seminorms (11). To see the bound also when all those seminorms vanish, apply continuity to every scalar multiple of a; then $La = 0$. All the finitely many tests lie in some $\mathcal{D}_m$, so $\ker(a \mapsto Ja|_{\mathcal{D}_m}) \subseteq \ker L$. This yields the claimed factorization through the m-dimensional cylinder dual. □

Remark 11 (The choice of interface is substantive). The topology τ_J is the weakest locally convex topology making the specified test pairings continuous. The completion theorem is a classical completion principle applied to an explicit dense arithmetic map. It does not say that $\mathcal{D}'$ is initial among all action-preserving carriers, nor that δ_1 is forced by the actions: every unit-supported vacuum in Proposition 8 is allowed. For example δ_u for $u \in \widehat{\mathbb{Z}}^\times$ yields another map. Requiring the integer-point origin fixes δ_1 as additional data. Nor do the transpose actions define a positive or signed Weil form.

6 Prior art, receipts, and the remaining boundary

Finite doubly commuting pure isometries admit the standard wandering-space model; Sarkar's Corollary 3.2 and Theorem 3.3 [1] explicitly give the finite-tuple decomposition and its multiplicity-space interpretation. Thus the mechanism behind the finite cases of Proposition 5 is classical. Dan–Guo's Definition 1.1 and Corollary 1.4 distinguish the additional Beurling-generation property for infinite sequences; their Proposition 3.4 and Remark 3.5 exhibit the failure of nonzero joint defect without that property [2]. No novelty is claimed for failure of a naive countable Wold extension. Bost–Connes [3] is adjacent arithmetic operator-algebra context, not a cited proof of the present selected-prime criterion or coefficient estimates.

The contributions new to this repository are the finite-family quantifiers, the sharp $d_F(n)$ shell estimate, the forward-only cylinder argument, and the explicit completion interface and its counterexamples. The methods are elementary CRT, geometric series, finite simultaneous diagonalization, density, and completion theory. The bounded literature check above does not establish scholarly priority for these specific refinements; that status is **UNVERIFIED**.

The independent module `observational_succ/forge_operator_transport.py` and its seven tests enumerate rational congruence cylinders, check selected valuations against direct counts, detect an unselected-factor normalization error, check permutation failures at primes dividing the test modulus, verify finite geometric orbit masses, and construct exact finite distribution interpolants. These checks do not prove reciprocal divergence, WOT convergence, Hilbert-space classification, or a completion property. Those are the mathematical arguments above.

The next discriminating work is a broader priority search for the forward-only disjointness theorem and a formal measure-theoretic verification of its density step. Neither is a missing premise of the written proofs. There is no inference from any result here to an identification with an original signed Weil form or to all-radius positivity. RH proved: **no**.

References

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